

Executive Summary — COVID-19 Geographic Distribution Project

- **Purpose:** Visualize and analyze the global geographic distribution of COVID-19 cases to reveal patterns, hotspots, and trends that inform public-health insights and resource prioritization.
- **Objective:** Aim to apply some of the techniques that include but are not limited to 1) Data cleaning 2) Data verification 3) Data exploration 4) Data aggregation 5) Data visualization

Possible manipulations: 1) Validate data 2) Use slicers by continent/month 3) Use geographical map to represent top 10 countries by deaths 4) Use geographical map to represent top 10 countries by deaths 5) Use pivot table (of cases and deaths by months (globally) and 6) answer research questions

- **Dataset:** Public COVID-19 cases data case dataset from an open source (e.g. on [Kaggle.com](https://www.kaggle.com/datasets/covid19/covid19) or <https://data.europa.eu/euodp/en/data/dataset/covid-19-coronavirus-data>) containing country-level daily confirmed cases, deaths, and location fields (country, continent).

The background of the project

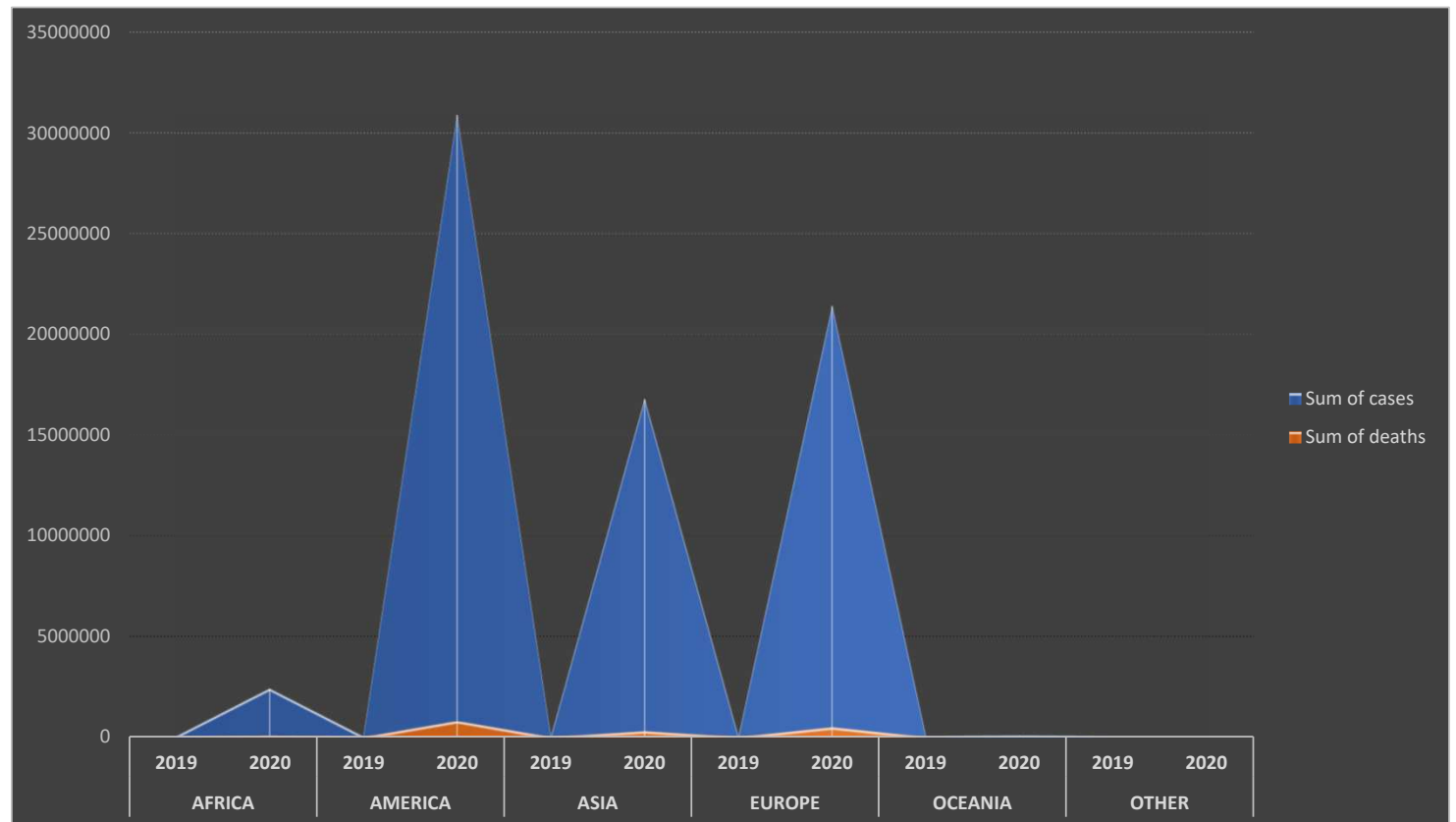
- **Problem context:** COVID-19 briefly state the pandemic's global impact and understanding geographic spread is essential for targeted response.
- **Motivation:** Publicly available time series data enable visual analyses that reveal where and when cases concentrated.
- **Data source:** COVID-19 dataset — daily time series of confirmed cases, deaths, and location (country, continent where available).

This analysis uses the EU open dataset COVID-19 time series to map and quantify how the pandemic unfolded geographically, supporting the outcomes and recommendations on the next slide."

The research questions:

- 1. Do the curves of cases and deaths have an upward trend globally?

Continent	Sum of cases	Sum of deaths
Africa	2380477	56334
2019	0	0
2020	2380477	56334
America	30912097	785420
2019	0	0
2020	30912097	785420
Asia	16782266	291015
2019	27	0
2020	16782239	291015
Europe	21407342	483733
2019	0	0
2020	21407342	483733
Oceania	53440	1154
2019	0	0
2020	53440	1154
Other	714	7
2019	0	0
2020	714	7
Grand Total	71536336	1617663



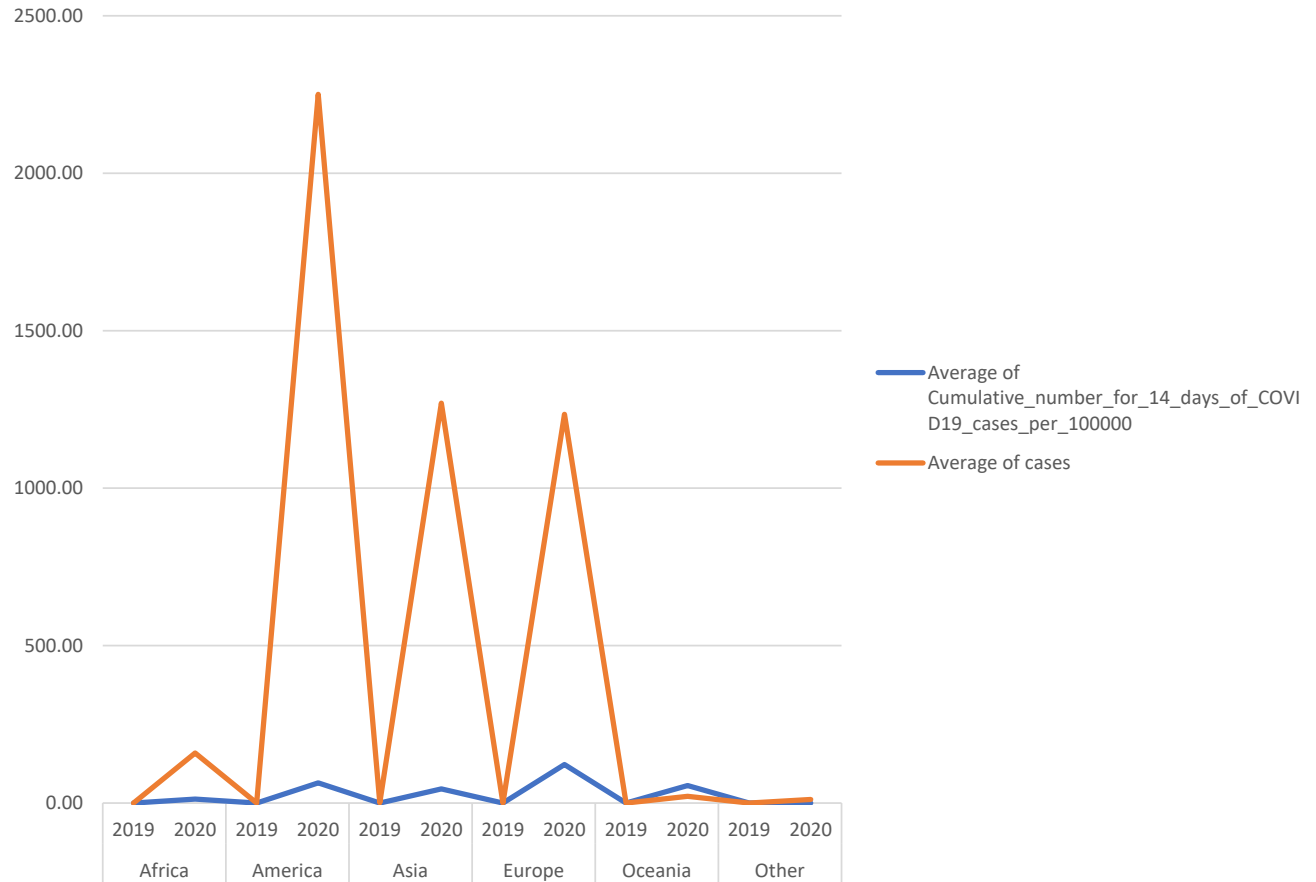
- 2. Does any continent have distinct ascending curve of cases?

-: I analyzed daily new confirmed COVID-19 cases and cumulative number for 14 days COVID-19 cases by continent for 2020.

Summary:

- 1) Clear sustained ascending curves (multi-week increases) occurred in 2020 for Europe, America.
- 2) Less distinct ascending patterns occurred at Asia, Africa, Oceania.

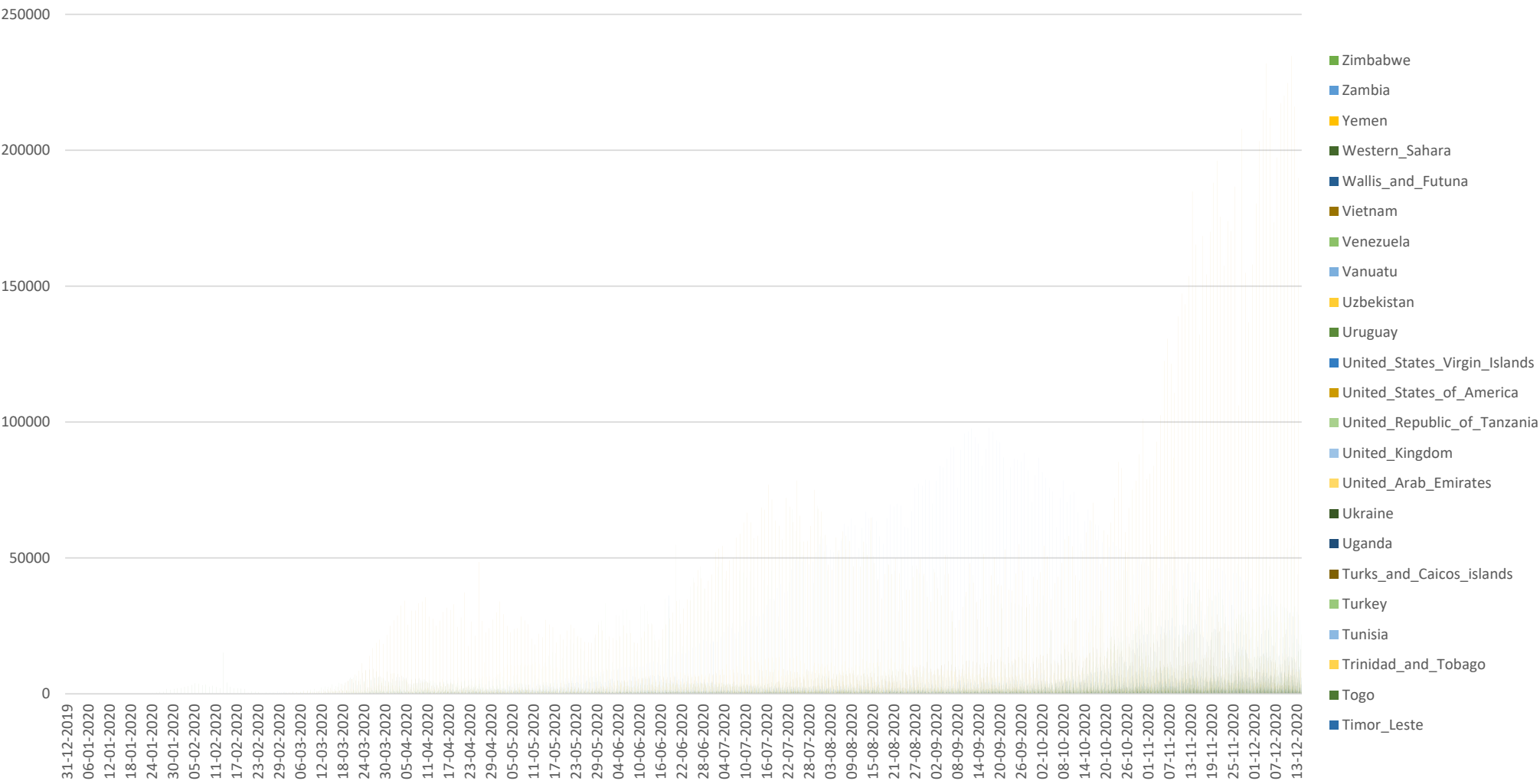
Year and continent	Average of Cumulative_number_for_14_days_of_COVID19_cases_per_100000	Average of cases
Africa	12.50	158.90
2019	0.00	0.00
2020	12.50	158.93
America	63.92	2249.46
2019	0.00	0.00
2020	63.94	2250.44
Asia	44.67	1267.45
2019	0.00	1.08
2020	44.75	1269.84
Europe	122.18	1232.29
2019	0.00	0.00
2020	122.39	1234.42
Oceania	55.48	21.38
2019	0.00	0.00
2020	55.53	21.39
Other	0.00	11.16
2019	0.00	0.00
2020	0.00	11.33
Grand Total	63.30	1155.68



3. What date was noted with the highest daily number of new cases? In which country?

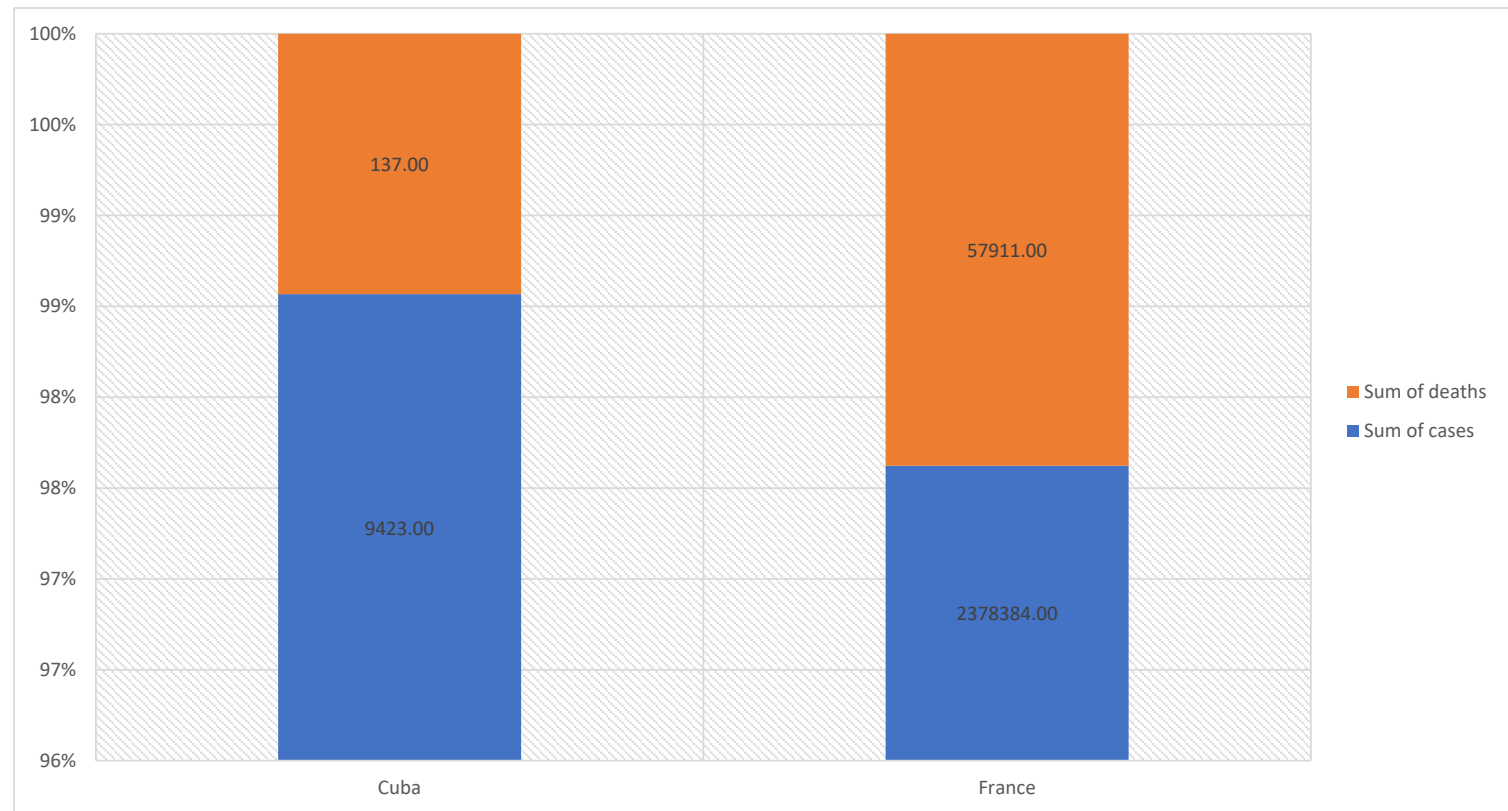
-: Series “ United_States_of_America” Point “12-12-2020”

Value: 234633



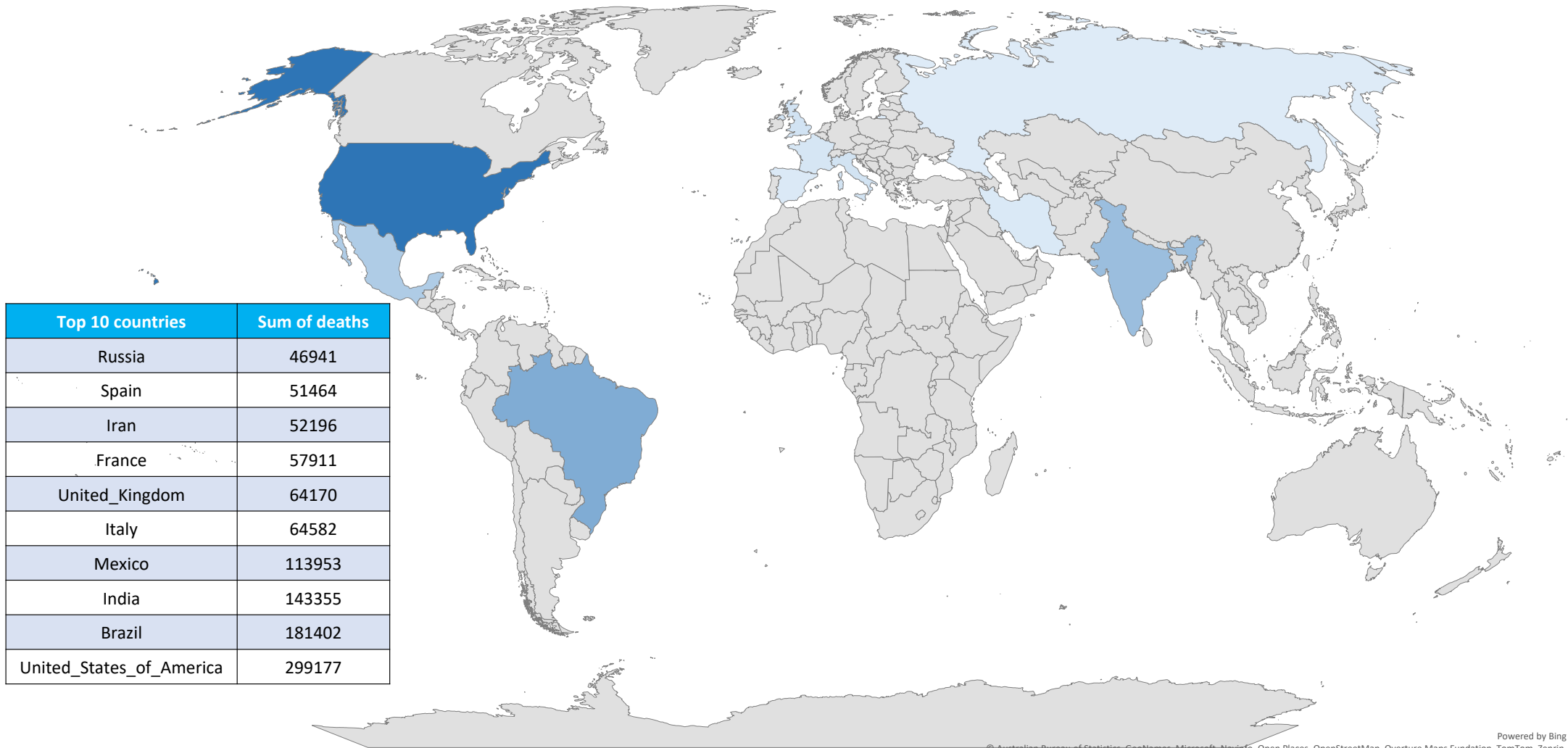
- 4. Does the epidemiological situation in France differs in comparison with Cuba ?
 - : Short comparison for yr 2019, 2020 (confirmed cases and deaths per million, epidemic curves, and testing.
 - 1) France's confirmed case rate was markedly higher than Cuba's.
 - 2) France had substantially higher COVID-19 mortality per capita.

Countries	Sum of cases	Sum of deaths
Cuba	9423.00	137.00
France	2378384.00	57911.00
Grand Total	2387807.00	58048.00



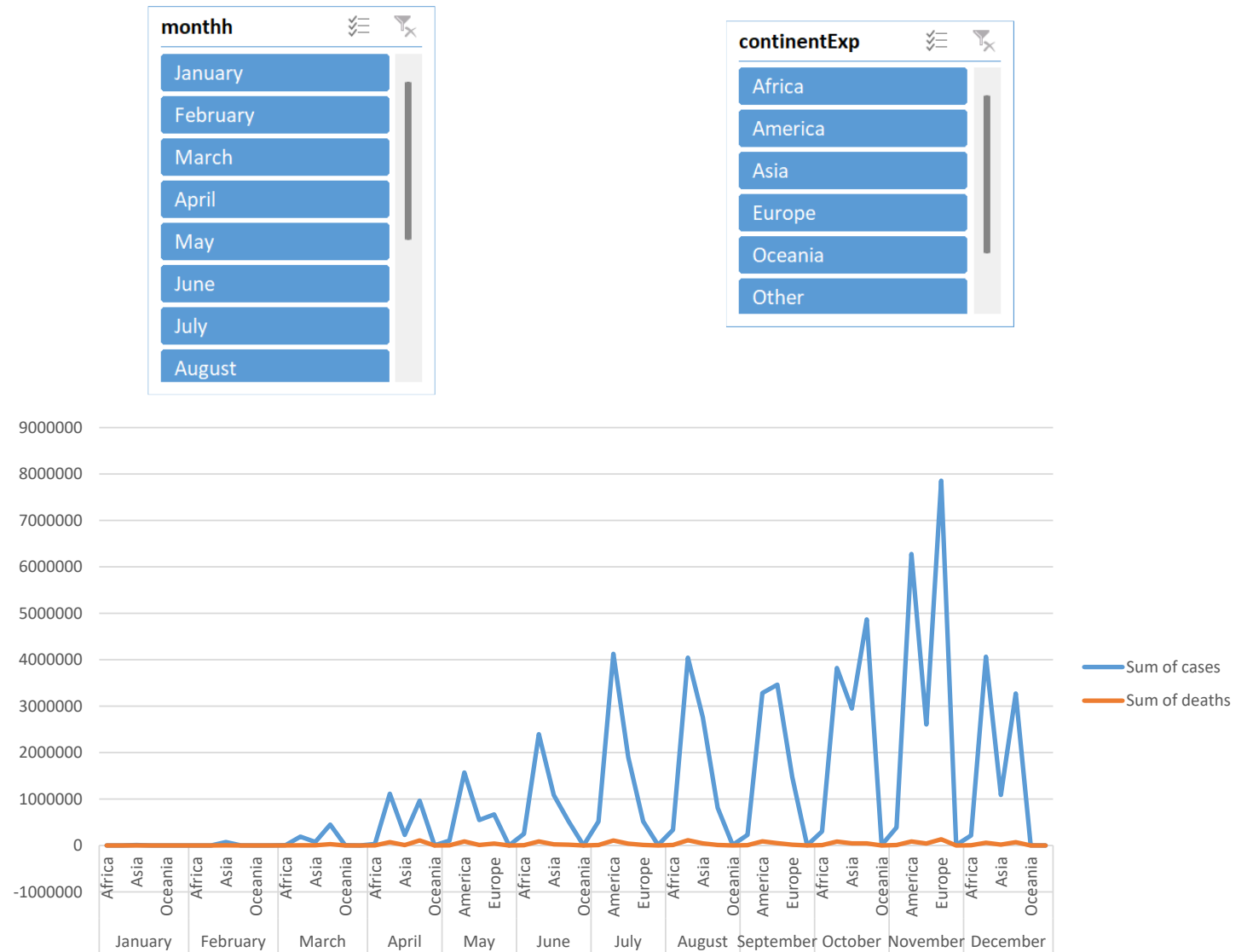
Top 10 countries by death

Sum of deaths
46941 299177



Use slicers by continent/month and Use pivot table (of cases and deaths by months (globally)).

Month	Sum of cases	Sum of deaths
January	9799	213
Africa	0	0
America	11	0
Asia	9766	213
Europe	15	0
Oceania	7	0
Other	0	0
February	75422	2708
Africa	3	0
America	86	0
Asia	73470	2679
Europe	1139	23
Oceania	19	0
Other	705	6
March	723738	35814
Africa	5134	166
America	190000	3687
Asia	75044	3822
Europe	448271	28117
Oceania	5298	21
Other	-9	1
April	2339594	189180
Africa	31598	1425
America	1113933	70904
Asia	227059	8408
Europe	964192	108348
Oceania	2812	95
Grand Total	3148553	227915



- Steps taken to clean the data:-

- ✓ Data cleaning
- ✓ Data verification
- ✓ Data exploration
- ✓ Data aggregation
- ✓ Data visualization

- The approach/method/tool used:-

- ✓ Excel
- ✓ Power BI
- ✓ Power query